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PERTURBATION MODELING AND TRAJECTORY DISPERSION ANALYSIS FOR THE IM-2 NOVA-C MISSION

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The IM-2 Athena spacecraft has been designed to support precision landing within a 50-meter landing radius at Mons Mouton located 150 km from the lunar South Pole. Updates to the maneuver execution error model for the spacecraft's cryogenic liquid oxygen/methane main engine have been implemented based on post-flight performance analysis from the IM-1 Odysseus mission. The "non-propulsive" thermal vent system jets have been redesigned and repositioned to mitigate the disruptive forces and torques that affected IM-1. In addition, new on-orbit test events are planned for RCS jet forces characterization, star tracker to IMU misalignment observation, and tail-off Δv compensation. A new strategy for burn planning and onboard burn execution has been developed with aim to minimize maneuver execution error and disturbances from small vent forces before and after the burns. The resulting pre-flight trajectory dispersion analysis for the IM-2 mission from launch vehicle separation to initiation of terminal guidance is presented in this paper. This preliminary analysis has led to the development of updated maneuver design strategies, which demonstrate promising results and instill confidence in the anticipated success of IM-2.

INTRODUCTION

Pre-mission analysis of the nominal trajectory dispersion corridor for IM-1 assumed that maneuver execution errors would be predominantly driven by sensed acceleration errors during the execution of the maneuver.[1] While this is a key contributor, the impact of unmodeled accelerations related to main engine firing and during coasting flight resulted in dispersions significantly beyond the 3σ expectation for the delivery accuracy into Low Lunar Orbit (LLO) and De-orbit Insertion (DOI) for IM-1.[2] For IM-2, an updated maneuver execution error model was developed from a thorough characterization of the IM-2 guidance, navigation, control (GNC), and propulsion systems. The maneuver execution errors were implemented using a Gates model,[3] to more comprehensively capture the effects of all contributors associated with operating a cryogenic liquid oxygen/methane main engine in deep space. The improved execution error modeling, understanding of propulsive activities during coasting flight, and analysis of orbit prediction uncertainty are planned updates to the IM-2 dispersion analysis results discussed in this paper.

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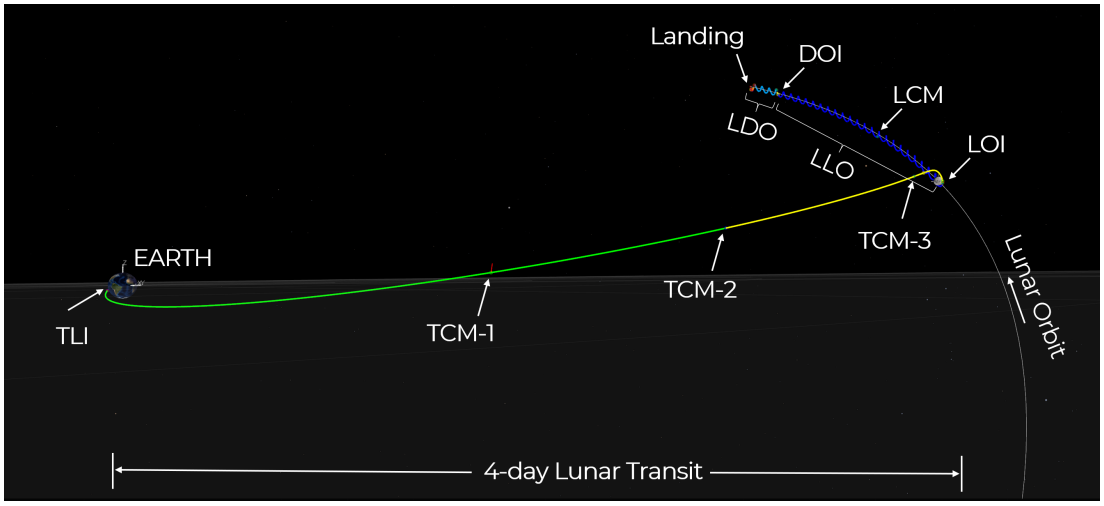


Figure 1: IM-2 lunar mission design reference trajectory.

IM-2 MISSION DESIGN

The IM-2 trajectory design is an extension of the design executed for the IM-1 lunar mission. The launch is conducted from a Space-X Falcon-9 rocket, with Trans-lunar Injection (TLI) performed by the Falcon-9 upper stage. After launch vehicle separation, the Nova-C is delivered into approximately a 200 x 372,000 km orbit about the Earth on intercept trajectory for lunar rendezvous approximately 4.5 days later. During lunar transit, the Nova-C spacecraft has the opportunity to execute up to three Trajectory Correction Maneuvers (TCMs) to correct the launch dispersion and refine the lunar arrival condition. A co-planar, single burn Lunar Orbit Insertion (LOI) is used to capture into a 100x100 km LLO. Two architecture changes are incorporated in the IM-2 strategy based on IM-1 lessons learned. First, a statistical Lunar Correction Maneuver (LCM) is incorporated into the schedule to correct the LLO condition prior to DOI. For IM-1, the LCM maneuvers were not planned nominally; post-LOI dispersions were expected to be corrected with the DOI burn. Secondly, the post-DOI 100x10 km Lunar Descent Orbit (LDO) was extended to three full orbits for onboard Terrain Relative Navigation (TRN) initialization prior to Powered Descent Initiation (PDI). The entire surface-to-surface transfer from the Earth to the lunar surface takes about 7 days, with approximately 4.5 days in lunar transit, 2-3 days in LLO, six hours in LDO, ending finally with a 15-minute powered descent and final approach to the Intended Landing Site (ILS) at the lunar South Pole region.

NOVA-C PERTURBATIONS

Maneuver Error Modeling for IM-1

All burns prior to PDI were planned as finite burns with inertially fixed pointing. They were executed using the main engine with an inertial measurement unit (IMU) measuring the non-gravitational acceleration experienced by the spacecraft using accelerometers and gyroscopes. Immediately prior to main engine ignition, the aft-facing reaction control system (RCS) jets were fired for 8-10 seconds in an RCS settling burn to shift propellant towards the main engine. The expected Δv from this RCS settling was offset from the Δv that guidance aimed to achieve from the main engine. The pre-mission predictions for maneuver execution performance were assumed to be predominantly

driven by the accuracy of sensed errors from the IMU feedback. These were modeled as axial and lateral maneuver error components due to unavoidable errors in sensed vehicle attitude and acceleration during the burn. IM-1 post-flight analysis showed that other contributors to maneuver execution performance (including thrust vector control (TVC) misalignment, spacecraft center-of-mass knowledge, star tracker to IMU misalignment, and post-engine shutdown transients) were significant and needed to be measured, modeled, and accounted for more accurately for IM-2.[4]

Maneuver Error Modeling for IM-2

The Nova-C maneuver execution errors are assessed relative to the nominal, commanded maneuver, which assumes perfect performance. The Gates Model lays out four independent sources of error that contribute to the total error caused by the imperfect execution of the maneuver.[3] Maneuver execution error models generally have a *fixed* error that is independent of maneuver magnitude and a *proportional* error that scales with Δv. These proportional and fixed errors are further broken down into axial (along the commanded Δv vector) and lateral (perpendicular to the commanded Δv vector) directions with respect to the nominally commanded Δv vector. The axial errors are captured as components of: *Proportional Axial Error* and *Fixed Axial Error*, while the lateral errors are captured in components of *Proportional Lateral Error* and *Fixed Lateral Error*.

These errors arise from multiple sources, and analytically characterizing all potential contributors is time-intensive. As a result, a statistical approach is employed to efficiently address the complexity of the error sources. This involves running Monte Carlo simulations with varying input dispersions and a range of commanded burns, followed by an analysis of the accumulated Δv statistics from the outputs. This is discussed in further detail in the Simulation Modeling section. Equations 1 and 2 show how the total axial and lateral maneuver execution errors are derived by evaluating the actual perturbed Δv vector relative to the commanded Δv vector.

Total axial error magnitude from each Monte Carlo sample is given by:

$$\Delta v_{err\parallel} = \overline{\Delta v}_{act} \cdot \widehat{\Delta v}_{cmd} - ||\overline{\Delta v}_{cmd}|| \tag{1}$$

Total lateral error magnitude from each Monte Carlo sample is given by:

$$\Delta v_{err\perp} = ||\overline{\Delta v}_{act} - (\overline{\Delta v}_{act} \cdot \widehat{\Delta v}_{cmd}) * \widehat{\Delta v}_{cmd}|| \tag{2}$$

The standard deviation (σ) of the axial and lateral errors from each dataset provide the expected 1σ error associated with the corresponding commanded Δv magnitude. With a sufficient number of datasets covering the expected range of commanded Δv, a statistical model can be developed to predict the expected error. This model is based on a linear curve fit that relates each commanded Δv to the corresponding errors. In some cases, it may be necessary to partition the data into multiple curves to account for distinct error regimes that reflect unique performance behaviors of Δv. A representative case of a TCM lateral error model is illustrated in Figure 2, where the slope of the line represents the proportional lateral error and the y-intercept represents the fixed lateral error.

Propulsion System Updates

The IM-2 spacecraft can perform several different types of maneuvers using the main engine (ME) or aft-facing RCS jets. The effective Δv of a minimum impulse ME ignition or RCS thruster firing depends on the total mass of the spacecraft. For the Nova-C mission, minimum impulse

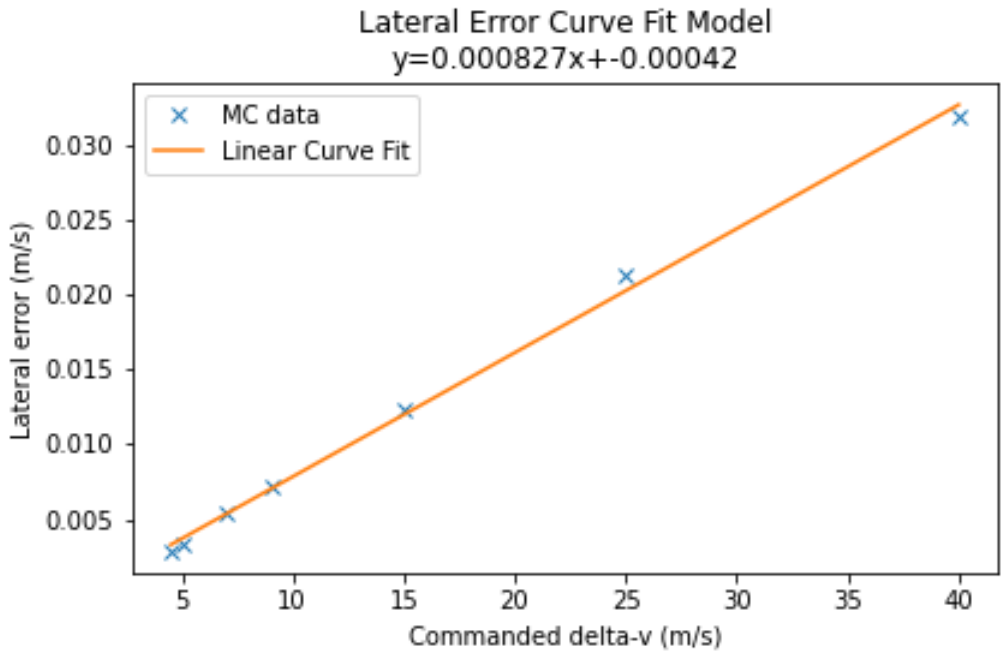


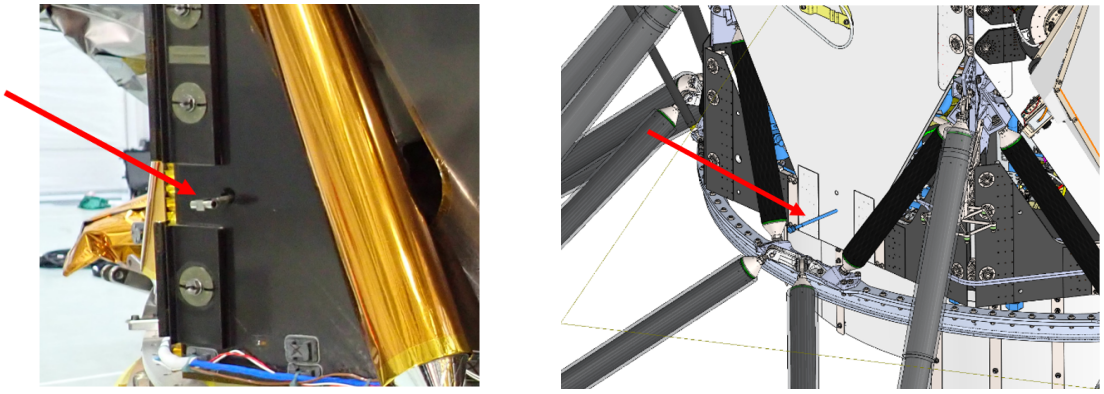
Figure 2: Lateral error model for TCM

maneuvers are only required during translunar coast prior to LOI. Thus, the effective minimum impulse Δv is discussed in the context of the full wet mass Nova-C vehicle configuration. RCS-only Δv maneuvers can be executed for Δv impulses less than 0.5 m/s. However, this requires a significant fraction of the spacecraft's available helium budget, and thus is only considered for the TCM-3 maneuver immediately preceding LOI, where the hyperbolic approach trajectory is most sensitive to small velocity dispersions. The RCS thrust level is also below the noise floor of the IMU and thus must be commanded with open-loop shutdown timing.

In contrast, the IM-2 spacecraft can execute a minimum Δv of 1.2 m/s with an abbreviated main engine ignition at 35% maximum thrust. However, this minimum Δv configuration cannot use closed-loop guidance for thrust control because the thrust impulse accumulates so quickly that the minimum Δv is achieved before guidance flight software (FSW) has sufficient time to run and manage the shutdown timing when running at a 5 Hz app rate. A minimum Δv of 1.8 m/s is achievable with open-loop guidance with a main engine ignition at 45% maximum thrust. Above Δv of 4 m/s, closed-loop guidance can be used to shut down the main engine based on sensed acceleration. For Δv between 1.8 and 4.0 m/s, guidance uses a pre-computed shutdown timing table to achieve precise maneuver execution performance.

In preparation for a main engine burn, a feedline prep process must be performed to condition the propellant temperatures for ignition. This process results in the venting of liquid oxygen (LOX) and liquid methane (LCH4) prior to the burn. In addition, a feedline-inert process is also required after the main engine is shutdown, resulting in venting after each burn. For IM-1, feedline prep and inerting was initially performed using T-vents for the LOX and methane lines. Each T-vent is a single vent line that has two openings in opposing lateral directions located to minimize vehicle self impingement from the flow expansion. During the mission, mission operators observed that this T-

vent design generated significantly larger forces and torques than expected. After the first maneuver during IM-1, mission operators elected to vent the cryogenic propellants out of the straight tank fill pipes instead of the thermal vent system (TVS) because it was shown to produce less perturbing torque on the spacecraft. For IM-2, a new design for non-propulsive vents (NPVs) will be utilized which are expected to produce an order of magnitude smaller disturbance forces and torques on the spacecraft during feedline prep and inerting around each burn. Figure 3 shows the placement of TVS vents on the IM-1 and IM-2 spacecraft.



(a) IM-1 T-vent Geometry

(b) IM-2 Non-Propulsive Vent Geometry

Figure 3: Placement of the TVS vent on the IM-1 spacecraft resulted in unbalanced plume impingement directly against primary structure, which produced a sustained disturbance torque when active. For IM-2, the vent is extended to minimize structural impingement, and located such that residual impingements produce less net torque. During IM-1, the TVS was estimated to have produced 0.5N force constantly when active. For comparison, each RCS jet firing produces 2.5N thrust for 40ms. Had this TVS system been used for all burns, it is estimated that 1.3 kg of RCS helium would have been required to counteract the resulting disturbance. The TVS changes for IM-2 are predicted to reduce this net RCS usage required to counteract the TVS disturbance by 80% to 280g.

Feedline inerting occurs after burn completion. The LOX and LCH4 propellant lines contain trapped and unused cryogenic propellant after main engine shutdown, which gradually warms up and must be vented as the line pressure increases. For IM-1, this unused propellant was vented through the tank fill lines at the launch vehicle interface at the bottom of the spacecraft. The LOX and LCH4 lines are located on opposing sides of the spacecraft center of mass. However, since the LOX and LCH4 venting is not guaranteed to occur at the same time (process occurs asynchronously over 10-20 minutes), the feedline venting produced equal and opposite torques that needed to be nullified by RCS to maintain spacecraft pointing after the burn. For IM-2, the feedline inerting will be conducted through the main engine. The spacecraft will be held in burn attitude for 20 minutes after the burn, and the main engine gimbals are commanded to continue pointing through the vehicle center of mass while the feedline inerting process is executed. Thus, a net acceleration (with negligible induced torque) will be achieved from feedline inerting in the desired Δv direction, which can be offset in the Δv command that is targeted at main engine shutdown.

In addition to vehicle venting perturbations, there was also a substantial tail-off error observed during IM-1. Based on lessons learned from post-flight analysis of the IM-1 mission, adjustments in the the automated shutdown timing are expected to reduce this error from 2 m/s (observed during

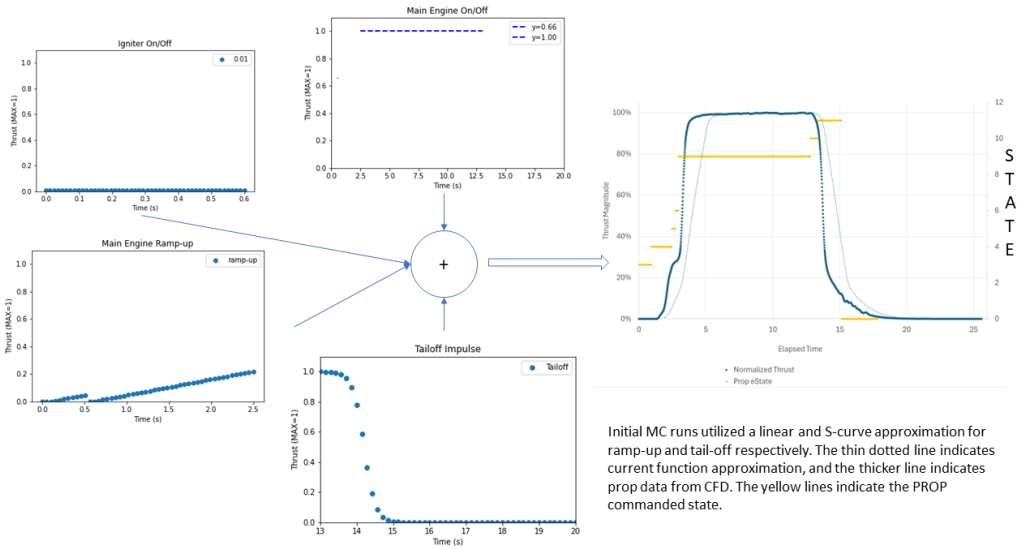


Figure 4: Comparison of the main engine thrust profile between the 6DOF simulation versus CFD

IM-1) to less than 0.1 m/s for IM-2.[2]

Simulation Modeling Using a high-fidelity 6 degrees-of-freedom (6DOF) spacecraft dynamics simulation, detailed modeling of the mass properties, fuel slosh, main engine gimbal actuators, main engine thrust, TVS venting, and RCS jet firings for various commanded maneuvers are run through a Monte Carlo (MC) to create thousands of outputs for total cumulative Δv through a burn. The onboard GNC flight software is run in the loop, commanding the spacecraft effectors while monitoring feedback from available navigation sensors.

The engine model in the 6DOF simulation utilizes a series of simple functions to model the thrust ramp up during ignition and ramp down after shutdown as a function of target power level. Figure 3 shows how the model is tuned to match the expected thrust from computational fluid dynamics (CFD) analysis. The ramp-up and tail-off impulse functions are described by linear and inverse exponential (S-curve) approximations. Figure 4 shows the model normalized to a maximum thrust value of one. These functions enable engine performance to be dispersed for MC runs as a function of target power level and FSW configuration.

Dispersions for the vehicle uncertainties are modeled to a nominal 3σ level and thus the observed Δv statistics can be compared to the commanded target Δv to inform the maneuver execution error model used in the trajectory dispersion analysis. By evaluating the accumulated Δv for a set of n MC cases for each of these regimes, a linear fit in both the axial and perpendicular directions can be derived to best capture the error model for any given Δv. Comparing the total accumulated Δv for each of these sets with the nominal perfect navigation case produces the standard deviation for the error in both the axial and the perpendicular directions. Performing a least squares fit for the axial and perpendicular errors respectively can generate the four parameters required in the Gates model. This process is repeated for Δv regimes for small burns (1.2-4 m/s), medium burns (4-20 m/s), and large burns (830 m/s).

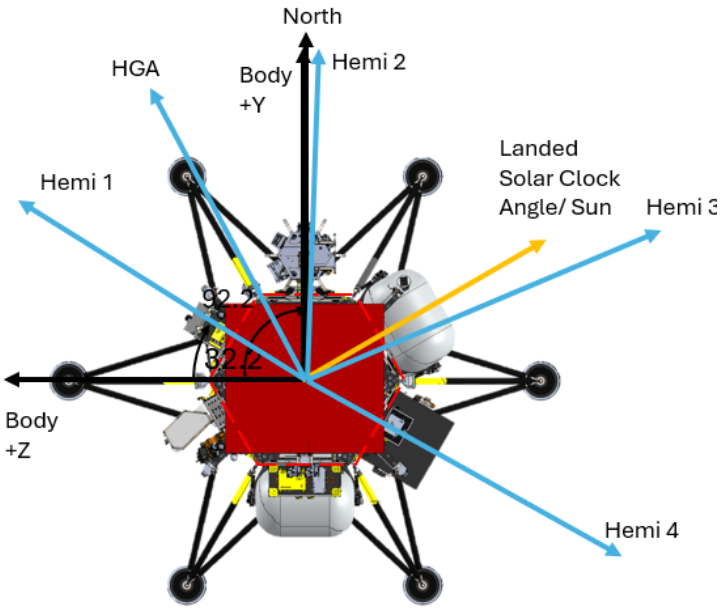


Figure 5: Top-down view of the IM-2 Nova-C spacecraft.

Other Perturbations

Attitude Timeline For the majority of flight during lunar transit and in lunar orbit, the spacecraft will be in a fixed pointing mode with hemispherical (Hemi) antennas pointing to Earth for communications (primary) and solar panels to the Sun (secondary). Figure 5 shows the antenna configuration on the IM-2 Nova-C spacecraft. At landing, the +Y side of the spacecraft will be oriented along the local meridian toward the North Pole. Hemi antennas one and two, as well as the high-gain antenna (HGA), are located on the +Y side of the spacecraft for communications with Earth from the lunar surface. The solar panels are also primarily on this side of the spacecraft for line of sight to the Sun during the landed surface mission. Hemi antennas three and four are centered on the -Z axis of the spacecraft. For the majority of the flight, the vehicle is oriented with -Z to Earth (primary) and +Y to Sun (secondary). TCMs 1-3, LOI, LCM, and DOI are all spaced approximately one day apart. For each of these maneuvers, the vehicle will rotate to maneuver attitude 20 minutes prior to the burn for feedline prep, execute the burn, hold attitude for 20 minutes for feedline inert. Once the feedline inerting hold is complete, the vehicle will rotate to HGA to Earth for high data rate data transfer to Earth, and then return to normal coast pointing.

The Nova-C vehicle uses RCS jets fired in antiparallel pairs to create a nearly pure torque. In a perfect nominal scenario, RCS attitude maneuvers will induce no translational thrust. However, the final as-built thrusters will include some nonzero jet-to-jet misalignments, and the expectation is that there will be a small translational force with each pair of jet firings. Differences in commanded versus actual torque are expected to be reduced after an initial RCS Ping Test checkout, where the actual thrust for each RCS jet is measured and the thrust values are updated accordingly in the onboard jet select logic.

RCS Attitude Adjustment Perturbations The RCS Ping Test for IM-2 will fire each jet independently for approximately 100 ms and then come to rest. Since the RCS thrust level is below the

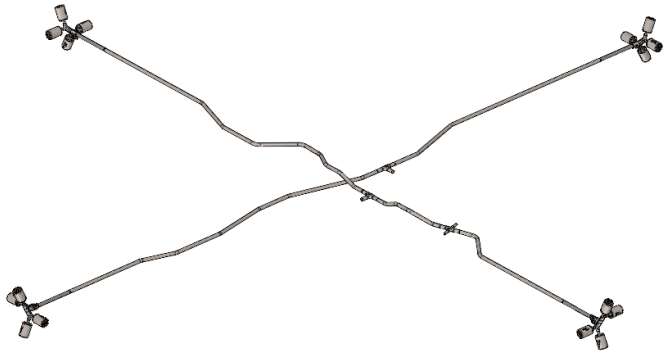


Figure 6: Nova-C RCS jet configuration with four RCS pods each hosting four thrusters.

noise floor of the IMU, the observed rotational response is used to estimate the effective orientation and thrust of each RCS jet. The RCS Ping Test will also reveal any possible changes to the expected perturbative forces. For IM-1, many of the perturbations were due to unexpected thermal issues which caused some RCS jet leakage at extremely low temperatures. These issues are expected to be addressed on the IM-2 Nova-C spacecraft, and hence the nominal mission for IM-2 is expected to have significantly lower RCS activity than the nominal IM-1 mission. At launch, the Nova-C is programmed to autonomously execute the nominal attitude timeline for the current Design Reference Mission (DRM). However, the spacecraft attitude timeline will be adjusted during the mission as needed for operations. Table 1 shows the estimated total number of seconds for RCS firings and the associated induced Δv based on the DRM.

Estimates of the RCS jet on times for each of the major rotational and translation maneuver events during the mission are characterized from 6DOF MC analysis of the IM-2 RCS dynamics around each event. The statistics for the expected RCS jet on times and resulting perturbational Δv's are shown in Table 1. Note that although RCS is used for settling burns, these perturbations will be factored into commanded Δv and thus should not substantially affect the maneuver execution error. Table 1 shows the total seconds of RCS on time around each maneuver type (excluding the settling burn interval for translational maneuvers).

Table 1: Preliminary IM-2 Mission RCS Perturbations

Mission Time (hrs)	Event	RCS Seconds	Expected Δv (m/s)
0	LV Separation	4.6	0.00028
4.3	Nav Camera Calibration	25.4	0.00684
24	TCM-1	97.5	0.0418
48	TCM-2	97.5	0.0418
90	TCM-3	97.5	0.0418
108	LOI	97.9	0.1808
126	LCM-1	84.66	0.06157
151	DOI	66.32	0.05293

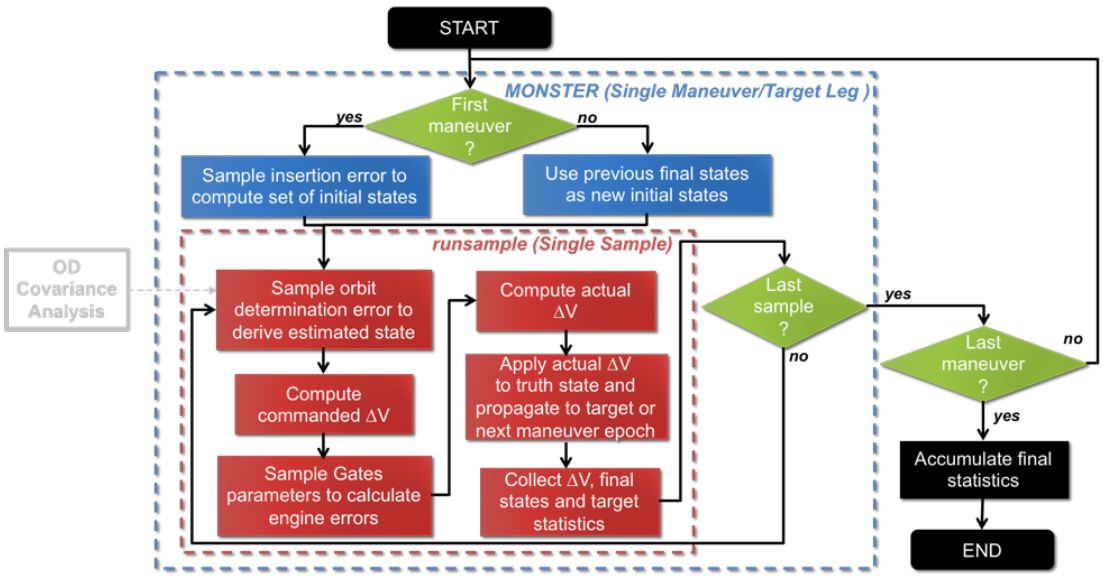


Figure 7: Overview of the Monte Carlo simulation and analysis process using the MONSTER (Monte Carlo Operational Navigation Simulation for Trajectory Evaluation and Research) software suite.

TRAJECTORY DISPERSION ANALYSIS

Monte Carlo Overview

Monte Carlo analysis is applied to the IM-2 mission to evaluate trajectory dispersions with respect to safety, delivery performance, and the ability to achieve mission objectives under known constraints, errors, and uncertainties. The impact of maneuver execution errors, along with other error sources such as orbit determination (OD) knowledge, is assessed using the MC framework outlined in Figure 7. The MC analysis incorporates a maneuver execution error model derived from IM-1 flight data[2] and historical mission experience. OD knowledge uncertainties were excluded from this dispersion analysis. The primary objective of this first-order analysis is to assess the robustness of the nominal mission plan against launch vehicle (LV) separation uncertainties and maneuver execution errors, while considering spacecraft system limitations. To account for the minimum Δv threshold, cases requiring less than this threshold to reach the target are skipped or adjusted, with Δv fixed at the minimum and target constraints allowed to vary.

The trajectory MC analysis begins just after LV separation, where the spacecraft is released from the Falcon 9 upper stage to continue on its way toward the moon. The initial truth state for each trajectory is sampled from a SpaceX-provided covariance encapsulating the expected performance of the LV. Without correction, these ballistic trajectories provide the Lunar B-Plane arrival conditions as shown in Figure 8. As stated previously, the IM-2 mission has three scheduled TCMs planned during the transit to the moon. TCM-1 is placed 18 hours after LV separation in order to remove as much of the injection error as possible. The contour shown in Figure 8 shows the range of possible TCM-1 Δv magnitudes required to correct for these errors, which must be accounted for in the mission Δv budget. Additional correction maneuvers, TCMs 2 and 3 are expected to provide smaller magnitude corrections further reducing the delivery error to the LOI state.

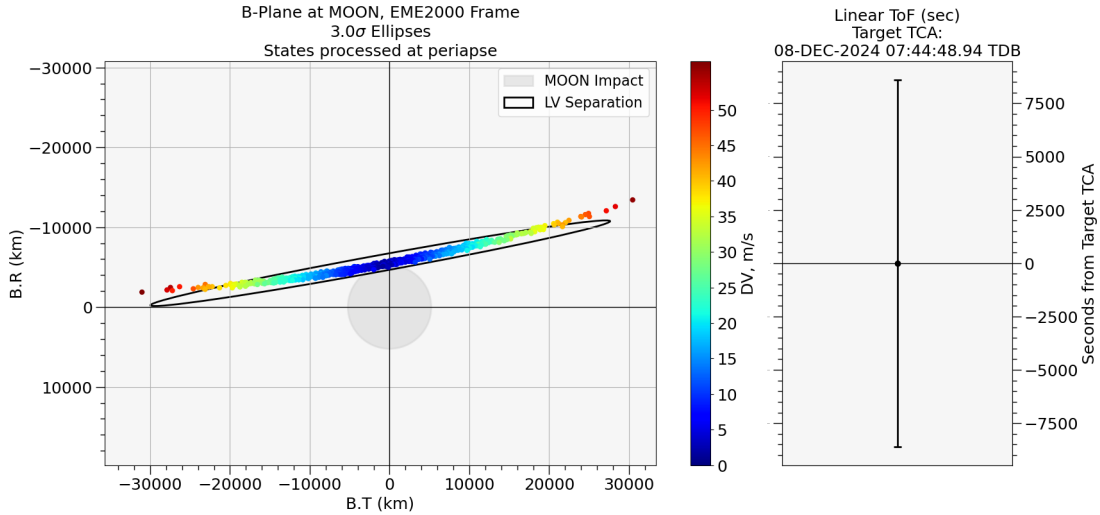


Figure 8: Launch vehicle separation B-Plane illustrating Δv cost for TCM-1 correction.

The spacecraft main engine is expected to support a wide range of Δv for the mission. The largest being LOI, nominally 829 m/s, while the correction maneuvers could be in the realm of centimeters per second. The lower limit Δv constraint has led to significant dispersion results when cases skip maneuvers that would otherwise be too small to execute. Figure 9 shows the delivery performance when skipping maneuvers that are below the min Δv capability of the spacecraft. 95.2% of TCM-2 cases are skipped in this scenario and 78.0% of TCM-3 cases are skipped, not allowing for any fine-tune correction leading up to the lunar orbit insertion. This scenario also shows that for TCM-3, a significant percentage of cases result in altitudes as low as 30 kilometers (3σ) near the time of orbit insertion. Several strategies are employed to improve this delivery performance.

Lunar Approach

The first trajectory correction maneuver, TCM-1, is nominally planned to execute 18 hours after LV separation. Preliminary analysis indicates that the TCM-1 Δv could range from sub-meter per second up to nearly 60 m/s in magnitude. Although the spacecraft can support maneuvers as small as 1.8 m/s, the analysis shown here assumes a flight rule requiring this first maneuver to be no less than 5 m/s. This would serve as an early opportunity to characterize the in-flight main engine performance and make adjustments, if needed, for subsequent approach maneuvers. The cases resulting in Δv less than 5 m/s to reach the LOI target were set to 5 m/s and allowed the time of lunar closest approach to vary, arriving early or late. Figure 10, shows the TCM-1 B-Plane where the larger ellipse represents 83.7% of the samples where the Δv was large enough to target the desired state as nominally planned. The TCM-1 nominal targeting ellipse is larger due to the proportional components of the execution error model activated by these larger burns. The remaining 16.3% needed less than 5 m/s to achieve the same target, therefore Δv commanded to exactly 5 m/s and adjusted the lunar time of close approach. The trade-off for the smaller TCM-1 fixed Δv ellipse, is the increased time of flight uncertainty. For these cases, the time of periapse was intentionally set to arrive early, indicated by the mean result shown in the center of the 3σ error bars of the linear time of flight subplot. A similar strategy was applied at each of the remaining TCMs, using 1.8 m/s instead of 5 m/s. The extreme LV separation cases correspond to the largest TCM-1 Δv cases and

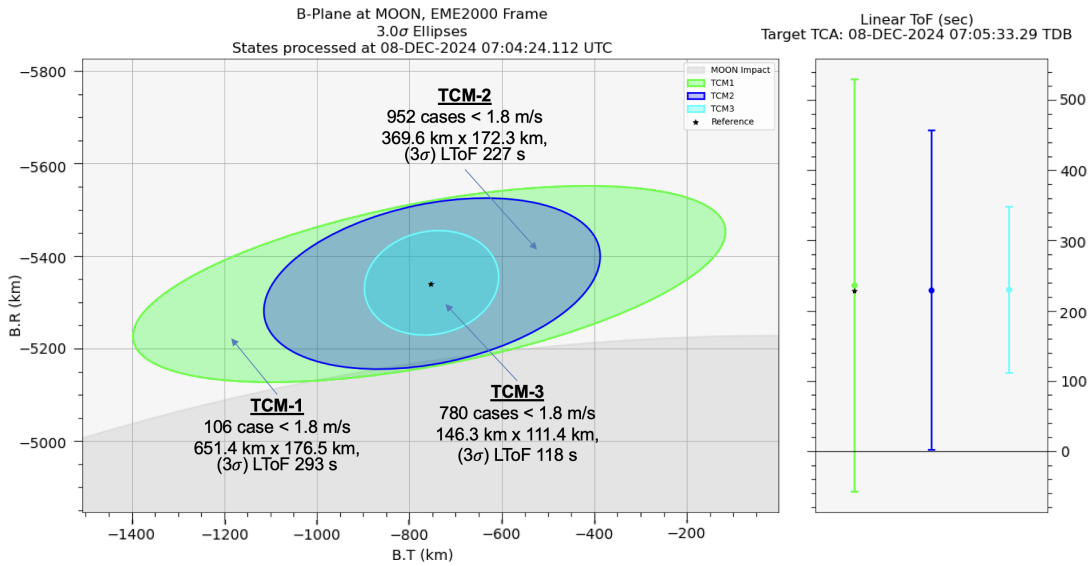


Figure 9: Resulting B-Plane when skipping maneuvers below the 1.8 m/s threshold.

mitigation strategies are explored at a later time.

The strategy described above for TCM-1 is slightly modified for TCMs 2 and 3. These subsequent correction opportunities may have up to three maneuver design possibilities.

1. Target the LOI insertion point nominally.

2. If the nominal targeting method results in a Δv below a minimum set threshold, the maneuver will be skipped.

3. If the nominal targeting method results in a Δv that is less than 1.8 m/s but more than the set threshold, the Δv will be commanded to 1.8 m/s and the time of close approach allowed to vary as needed.

This general approach leads to more opportunities to perform the correction maneuvers, thus reducing the size of the B-Plane delivery ellipse, and the side-effect of increasing the time of close approach uncertainty. Effects of the varying time of close approach can reduced by allowing LOI burn epoch to vary. This strategy effectively creates three families of results to be analyzed for both TCM-2 and TCM-3; nominal targeting family, skipped burn family and fixed Δv family.

The TCM-2 minimum threshold was set to 1.4 m/s, leading to 85.8% of samples skipping this burn. This has a two-fold effect: the majority of these skipped samples are likely to require a TCM-3 burn that is large enough to execute, and it also makes TCM-3 a critically required maneuver, particularly for the samples that had Δv just below 1.4 m/s. Figure 11 illustrates the criticality of TCM-3 for the skipped TCM-2 samples, while the remaining 14.2% of samples were able to apply a correction. Future analyses can adjust the 1.4 m/s skip Δv limit, reducing the size of the "Skipped TCM-2" ellipse by reducing the limit, noting that this will also impact the subsequent TCM-3 Δv magnitude. The goal for setting the TCM-2 skip Δv limit is to guide samples toward a mean TCM-3 magnitude that will result in the least amount of error. TCM-3 is the most critical of the lunar

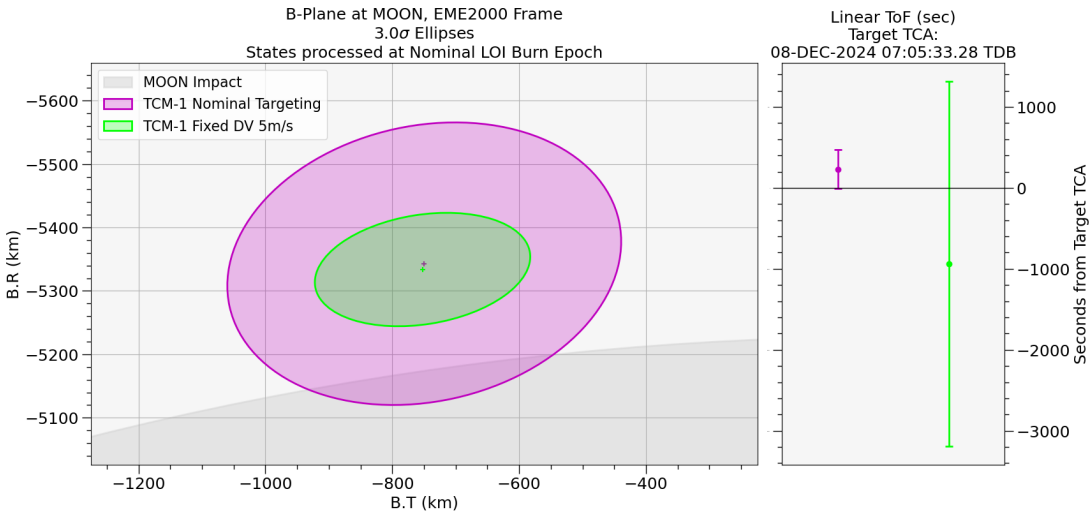


Figure 10: TCM-1 B-Plane near Lunar Orbit Insertion using preliminary lunar approach strategy.

approach maneuvers toward achieving a safe and desirable LLO, and should be steered toward a Δv magnitude that triggers the least amount of error.

TCM-3 occurs just 18 hours before LOI, and is the final opportunity to make a correction prior to orbit insertion. The goal for setting the TCM-3 skip Δv limit is to only skip TCM-3 if that skipped case is predicted to achieve an acceptable LOI state. A future implementation for this analysis will, rather than base the skip decision indirectly on a Δv magnitude limit, base it on a no-burn estimated target state error. In the end, the Δv limit was set to 12.5 cm/s for this study and only 0.4% of samples fell into that category. The majority of samples, 66.4%, fixed the TCM-3 Δv to 1.8 m/s, forcing the time of close approach to vary. The time of flight error for the fixed Δv cases is $\pm$121.4 seconds (3σ), to be mitigated with a varying LOI burn epoch.

The analysis shows that this strategy is effective, however, the values defined for the threshold of skip Δv is a tuning knob that can be re-calibrated in subsequent analyses that incorporate updated assumptions such as the execution error model and OD knowledge error. The resulting TCM-3 delivery to LOI state, in Figure 12, shows that by the time of TCM-3 the 99.6% of samples that perform TCM-3 have reduced the 3σ uncertainty to a level that ensures no samples will impact the surface and a lunar orbit can be subsequently achieved. The largest of the three ellipses, corresponding to the nominally targeted samples, is 17.5 km by 15.7 km (3σ).

Lunar Orbit

The nominal LLO is designed to be a circular 100 x 100 km altitude polar orbit. The delivery error to the LOI state affects how closely the design orbit can be achieved. Taking this into account, the targeting strategy for LOI applied in these analyses aimed to circularize LLO orbits when the incoming altitude, at the time of LOI execution, was equal to or below the nominal 146.3 km altitude. Cases that arrived above 146.3 km target a perilune altitude of 100 km, allowing deviation from circular. The 3σ radial delivery error at the nominal time of LOI is $\pm$51.8 km.

The orbital element statistics in Table 2, show that the LLO orbit is dispersed. These dispersions, if not corrected, leads to DOI 99th percentile Δv of 44.5 m/s and decreased delivery performance to

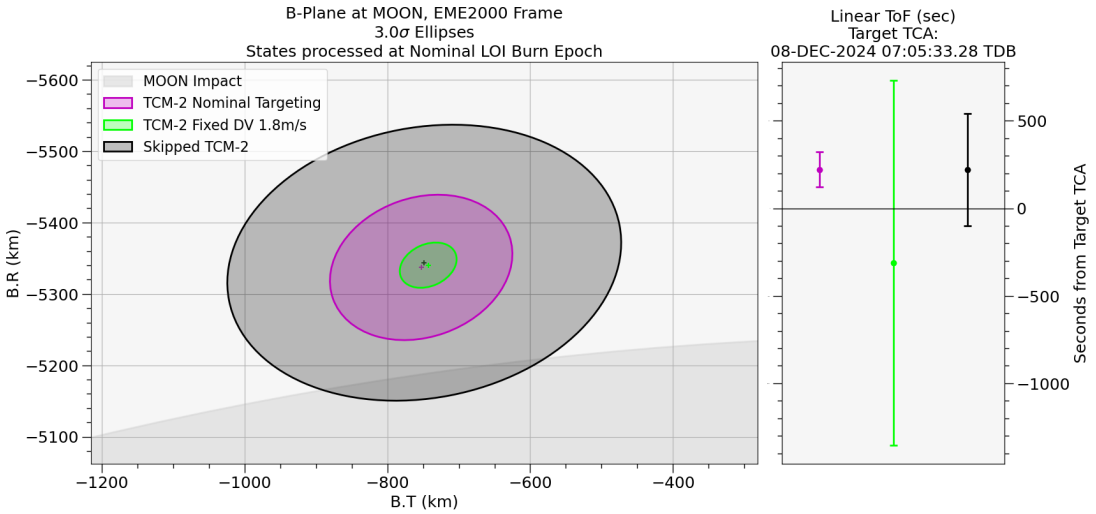


Figure 11: TCM-2 B-Plane near Lunar Orbit Insertion using preliminary Lunar approach strategy.

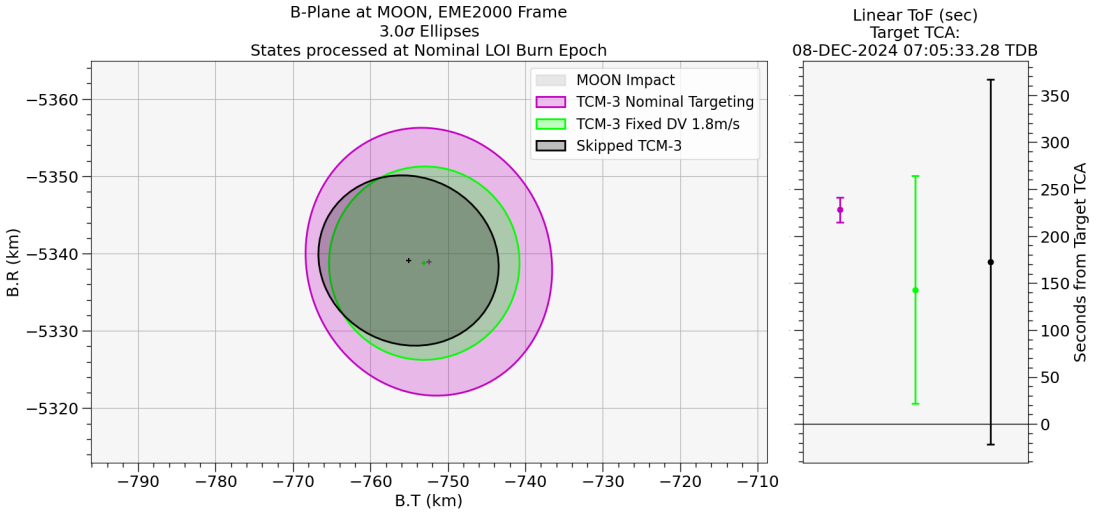


Figure 12: TCM-3 B-Plane near Lunar Orbit Insertion using preliminary Lunar approach strategy.

Table 2: Orbit element statistics 2-hours after LOI

	1%-tile	50%-tile	99%-tile
Perilune Altitude, km	58.78	91.45	113.15
Apolune Altitude, km	76.59	109.29	132.85
Inclination, deg	96.62	97.01	97.39
Argument of Periapse, deg	2.62	137.35	356.41
Right Ascension of Asc. Node, deg	70.48	70.77	71.05

the Braking Burn (BB) state, which marks the handover to the PDI phase. This prompted scheduling a statistical LCM to reduce dispersions before the DOI maneuver. For this analysis, the LCM burn start epoch nominally begins 24 hours after LOI and is allowed to vary ± 90 minutes. The targeting strategy applied for the LCM design includes the design of both LCM and DOI in conjunction to target the moon, body-fixed BB state. In this analysis, the post-LCM orbit was unconstrained, which led to some cases where the LCM initiated the LDO prematurely, as this solution minimized the overall Δv cost. Subsequent analyses will loosely constrain the post-LCM orbit such that the timing of this orbit transition will not deviate from the DRM trajectory.

This dispersion analysis concludes after the DOI burn is executed and the trajectory nominally transitions from LLO to LDO. The DOI maneuver, again, targets the same BB state and its start time is allowed to vary by ± 90 minutes. The targeted BB state epoch can also vary by ± 30 minutes to aid in Δv reduction and sample convergence. Table 3 provides the total Δv statistics for LOI, LCM, and DOI along the Lunar Orbit Totals row.

The trajectory culminates in the handover to PDI, which is best modeled with the high-fidelity, 6DOF simulator that models the autonomous spacecraft behavior down to the lunar surface. The final dispersions from the analysis described in this paper and shown in Table 4, the targeted PDI epoch was allowed to vary by ± 30 minutes, therefore these statistics were taken at the targeted PDI epoch of each individual sample. The final states from the DOI segment will become initial conditions for the 6DOF simulator MC analysis that applies the spacecraft FSW, system configuration and known perturbations of the lander down to the lunar surface. Statistically modeling the entire mission trajectory, from LV separation to surface landing, is essential for understanding final delivery performance, ensuring spacecraft and lander safety, and estimating the bounding propellant usage throughout the mission.

Conclusions

In summary, the dispersion analysis presented in this paper is considered preliminary as it provides insight to the statistical nature of the IM-2 Mission. Additional analysis will continue to be performed as new information is available. This preliminary analysis has led to a few early insights allowing for mission planners to prepare for operations.

- Minimum executable Δv: The minimum Δv floor of 1.8 m/s has a significant impact on the delivery performance to the moon and a mitigation strategy must be implemented.
- LV Separation: The LV separation covariance indicates the possibility of extreme cases where

TCM-1 Δv to recover cut deep into the mission Δv budget. Strategies to mitigate this will further be characterized in future work.

- Need for LCM: The delivery performance at LOI suggests that while there are no post-LOI safety concerns, the resulting orbit dispersions are sufficiently large to necessitate a clean-up maneuver. This maneuver is required to reduce the DOI Δv and subsequently improve the accuracy of the delivery to the BB state.

An important conclusion from dispersion analysis is to bound the overall mission Δv. Column "99%" in Table 3 provides the 99th percentile level of Δv allocated for each individual maneuver in the IM-2 pre-PDI mission plan, and the overall pre-PDI mission. The allocation for statistical Δv in excess of the deterministic requirement is highlighted at the bottom of the table, noting that there remains additional Δv allocated, separately from this table, for the spacecraft to achieve its autonomous descent to the lunar surface.

This analysis has been a productive step toward statistically characterizing the IM-2 Mission, yet continued analysis is still required. The execution error model described at the beginning of this paper will be a critical input to further analysis, as it will more accurately account for the expected performance of the spacecraft system. OD knowledge is another assumption that will improve the accuracy of statistical results; these analyses assume perfect OD knowledge. Some remaining perturbations that could also be accounted for in the next steps include stochastic accelerations and the

Table 3: Preliminary IM-2 Mission ΔV Statistics

Maneuver	Epoch [UTC]	Deterministic [m/s]	Mean [m/s]	SD [m/s]	1% [m/s]	99% [m/s]
TCM-1	04-Dec-24 06:35	0.414	15.12	10.02	4.88	46.93
TCM-2	05-Dec-24 06:35	-	0.77	2.13	0.00	9.36
TCM-3	07-Dec-24 13:04	-	2.21	1.25	1.73	5.86
LOI	08-Dec-24 07:04	828.89	829.63	3.12	822.66	837.31
LCM	09-Dec-24 08:00	-	6.40	4.44	0.00	16.00
DOI	10-Dec-24 13:39	21.20	21.12	5.28	11.00	34.74
Lunar Approach Totals		**0.414**	**18.11**	**10.31**	**6.71**	**48.97**
Lunar Orbit Totals		**850.09**	**857.11**	**7.52**	**841.31**	**876.16**
Pre-PDI Mission Totals		**850.50**	**875.26**	**12.65**	**850.92**	**908.42**
Allocation for Statistical Δv prior to PDI						74.00

Table 4: Preliminary IM-2 delivery error to the Braking Burn state extracted at each sample's targeted PDI epoch

Moon-Relative Position, km			Moon-Relative Velocity, km/s		
3σ Radial	3σ Transverse	3σ Normal	3σ Radial	3σ Transverse	3σ Normal
4.93E+00	4.48E+01	4.40E-01	5.64E-02	7.04E-03	2.73E-02

periodic propulsive venting events expected to occur while in lunar orbit. A final, pre-flight MC analysis that, at minimum, includes an IM-2 derived execution error model and OD knowledge derived from mission-specific covariance analysis will provide critical statistical information on what to expect from the latest mission plan. The targeting strategy for LCM must apply constraints preventing early departure from LLO. Next steps will also include improvements to the DOI targeting strategy to work in conjunction with the BB to target the 6DOF state of the hazard detection avoidance (HDA) activity for more realistic transition to the PDI phase. Additionally, MC analyses will continue to be performed during mission operations providing updated statistical predictions that reflect the latest operational scenario.

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